Hackathon Kickoff Session

MODELS TO MEDICINES

- Chairs: Xinyue “Clover” Chen & Tatiana Iakovleva
- Facilitator: Hongtao Yu
- Date: October 11, 2026

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Please note before we begin.

- **Views are personal**
  The views shared today are those of the individual speakers only.
- **No institutional representation**
  They do not represent any current or past institution or company.
- **Simulated data**
  The dataset is simulated. Results do not describe real patients or approved therapies.
- **Sponsorship**
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Introduction

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Hackathon Team



Xinyue “Clover" Chen

Co-chair

University of Florida
Graduate Assistant



Tatiana Iakovleva

Co-chair

University of Florida
Research Assistant Professor



Hongtao Yu

Facilitator

AstraZeneca

Co-chairs and facilitator are available to support you and will not participate in the final judgements

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Participant Team Formation

10 teams of 5 to 6 people, pre-assembled to be balanced.

Team 1	Team 2	Team 3	Team 4	Team 5
Team 6	Team 7	Team 8	Team 9	Team 10

Team name is optional

Please feel free to come up with your own team's name!

How teams were formed

- Mix of ML and covariate modeling expertise
- Mix of academia, industry, government, and non-profit
- Find your team number in the roster check

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Agenda for kick-off session

Sunday, October 11, 2026, 1:00 to 4:45 PM EDT

1:00-1:30	**Hackathon Logistics & the Challenge** Roster check, facilitator intros
1:30-1:45	**DM1 Clinical Background**
1:45-1:55	**DM1 Therapeutics**
1:55-2:05	*Break 1*
2:05-2:20	**Overall Workflow I** Data, Arms & Split
2:20-2:35	**Overall Workflow II** Pipeline & Deliverables
2:35-3:00	**Covariate Model Methodologies**
3:00-3:15	*Official coffee break*
3:15-3:40	**Machine Learning Model Methodologies**
3:40-3:55	**Assessment and Metrics**
3:55-4:10	**Q&A** Both tracks, with facilitators
4:10-4:45	**Team working time** **+ SharePoint Folder Access Confirmation**

Kick-off and background · Workflow · Methods · Assessment and Q&A · Breaks

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The Hackathon Challenge

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What are the participants' main tasks?

The Hackathon Question:
Can ML identify additional predictors and improve patient stratification beyond traditional covariate modeling?

- Teams will work with a synthetic clinical dataset, build predictive models, and present findings to expert judges.
- The main challenges:
  - ❑Be able to identify clinically meaningful covariates per treatment arm.
  - ❑Compare different covariate models and ML models with proper metrics.

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What Participants Received

Simulated Dataset

A simulated clinical trial dataset with 500 virtual patients covering all three treatment arms.

Methods Guide

A description of potential methods, spanning traditional covariate modeling and machine learning.

Tool Guidance

- Any platform, language, or software is welcome.
- AI tools are allowed with disclosure. The focus is meaningful covariate identification, not fancy models.

Dataset Shared at the Kickoff

Dataset and supporting materials shared during the kickoff session **through SharePoint.**

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Deliverables

Presentation

5-min talk + 3-min Q&A
(1 question from judges)
Wed Oct 14, 9:00 to 10:30 AM

Slide deck

The slides you present

Code files

All scripts behind your results
Any language or platform

Upload slides and code to your team SharePoint folder by the end of Tue Oct 13 (11:59 PM)

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How it will be evaluated

10 points

Clinical

- Key predictors per arm
- Real signal separated from noise
- Clinical interpretation and application

10 points

Technical

- Modeling workflow
- Unbiased split and justified method
- Assessment metrics

10 points

Communication

- Clear story within the time limit
- Supportive visuals
- Confident Q&A
- Meaningful conclusion

Meaningful covariate identification matters more than fancy models

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Prizes and recognition

Every participant

- Certificate of participation

Winning team

- Medals
- $100 Visa gift card

Honorable mentions

Possible for other teams, depending on the outcome

Winners are announced at the closing session on Wed Oct 14

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Hackathon Logistics

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Hackathon Day by Day Schedule

Kick-off

Sun Oct 11, 1:00–4:45 PM

- Share the dataset & materials with the participants.
- Orientation, background briefing, dataset walkthrough, and Q&A.

Working Period

Oct 11–13

- Teams work independently [Magnolia 1 available to use on Oct 12 & 13]
- Team upload presentation deck and code files to the SharePoint folder **by 11.59 pm ET of Tues Oct 13**

Please nominate a representative and send us an email saying the final version of the slides are ready.

Presentations

Wed Oct 14, 9:00–10:30 AM

- **10 teams (6-7 people)**
- 5-min team presentation plus 3-min Q&A [1 question]; judges evaluate live.

Awards

Wed Oct 14, 4:30–5:00 PM

Winners announced during the conference closing session.

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Office hours and routes of communication

Where to find files, help, and answers.

Team SharePoint folder

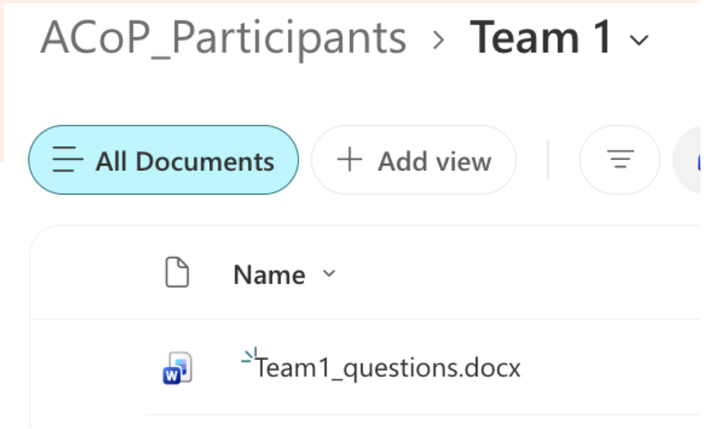
Every team member has access to a team-specific folder, e.g., "Team 1".
Use it to share files within your team and to submit your deliverables.

Office hour

Tue Oct 13, 8 to 9 AM, Magnolia 1.
Co-chairs and facilitator answer non-technical questions and give resource guidance.

Anytime during the conference

Upload any questions in your SharePoint folder
We will look at them and can address these during office hour.



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DM1 Clinical Background

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DM1 – Myotonic Dystrophy Type 1

A genetic, multisystem disorder characterized by progressive muscle weakness and myotonia, caused by an expansion of CTG repeats in the *DMPK* gene.

Genetic cause
Expansion of CTG repeats in the *DMPK* gene (on chromosome 19).

Key features
Progressive muscle weakness and **myotonia** (delayed muscle relaxation).

Multisystem involvement
Can affect the eyes, heart, respiratory system, endocrine system and more.

Variable course
Symptoms and severity can vary widely among individuals.

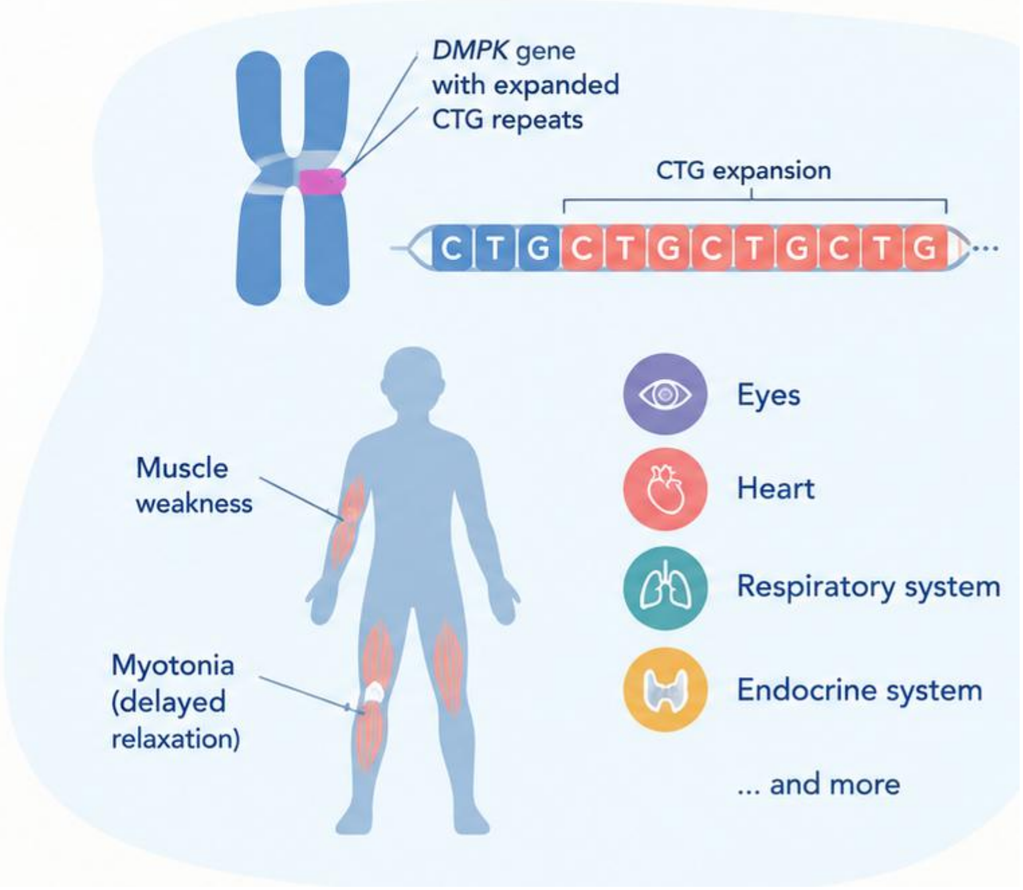


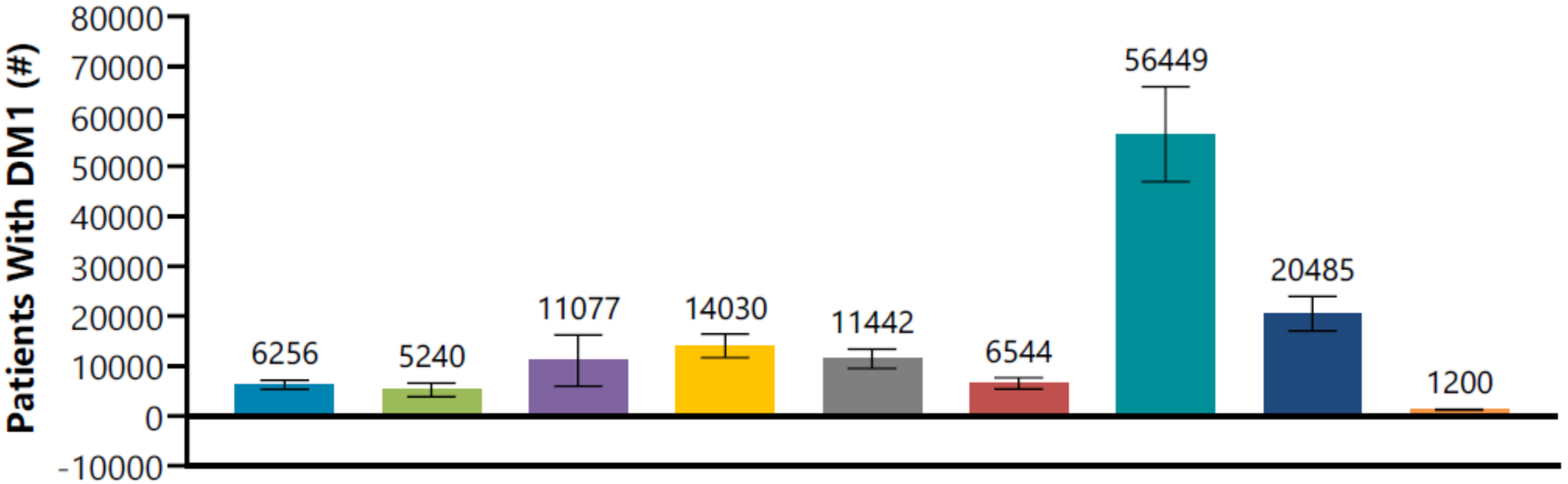
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DMPK gene: Dystrophia Myotonica Protein Kinase gene
CTG repeats: Cytosine –Thymine – Guanine repeats, a sequence of DNA nucleotides

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Rare by Rate, Substantial by Number

16.7 per 100,000, or 1 in 6,000 people



	UK	Italy	Spain	Germany	France	Canada	USA	Japan	Israel
Lower 95% CI	5,363	3,838	5,917	11,678	9,523	5,447	46,984	17,050	1,127
Upper 95% CI	7,150	6,641	16,189	16,382	13,360	7,641	65,913	23,920	1,268
Mean	6,256	5,240	11,077	14,030	11,442	6,544	56,449	20,485	1,200

Data are represented as mean±95% CI. DM1, myotonic dystrophy type 1.

Figure adapted from Rojas & Cong, ISPOR 2026 (EPH200).

Global Pooled

16.7

per 100,000

95% CI: 13.9 – 19.5

Highest Known

158/100K

Saguenay–Lac-Saint-Jean, Quebec, Canada

well-documented founder effect

Largest Population

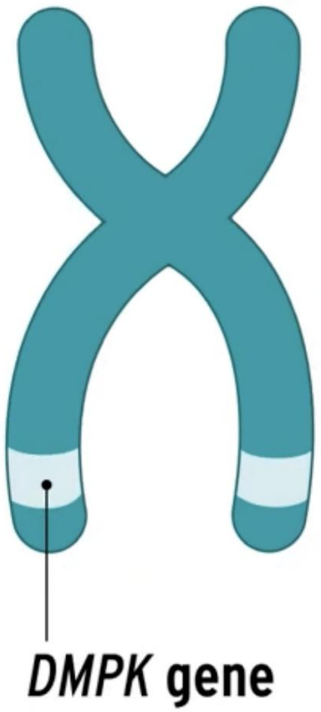
USA

~56,500 patients

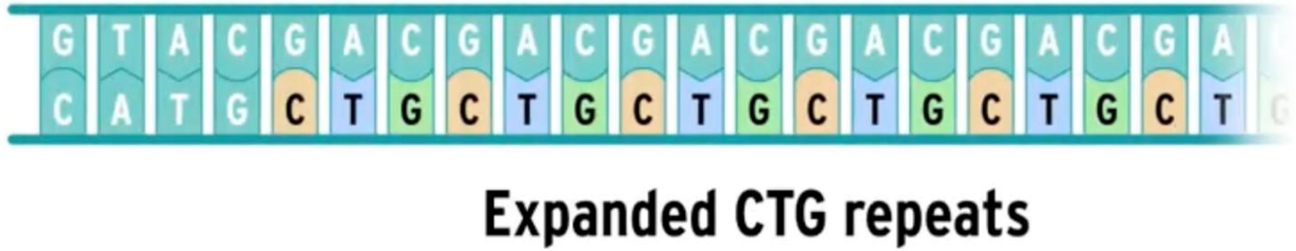
Note: No eligible publications identified for Germany, France, or USA; estimates derived via imputation from European data. Canada rate reflects imputed national estimate (excl. Quebec founder-effect region at 158/100K). Data: Rojas & Cong, ISPOR 2026. https://www.ispor.org/docs/default-source/cti-meeting-21463-documents/7fa04d15-1fbd-404c-a54d-67636d88ef5e.pdf?sfvrsn=5bb06370_0

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DM1 is caused by a mutation on chromosome 19, specifically within the DMPK gene



- Dystrophia myotonica protein kinase (DMPK) is not limited to one organ. It functions in muscle, heart, and brain cells. Additional roles have been identified in the thyroid, liver, pituitary gland, testis, and eye.
- DMPK gene has CTG repeats. In DM1, DMPK gene has expanded CTG repeats



Expanded CTG repeats

1.Myotonic Dystrophy Foundation. (2025). *Understanding DM1: From genetics to care* [Video]. YouTube. https://youtu.be/H2H2tr7jfag
2.The France Foundation. (2025). *Myotonic dystrophy type 1: Mechanism of disease* [Video]. YouTube. https://youtu.be/x_1dZlBXl88

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How a genetic mutation leads to multisystem dysfunction?

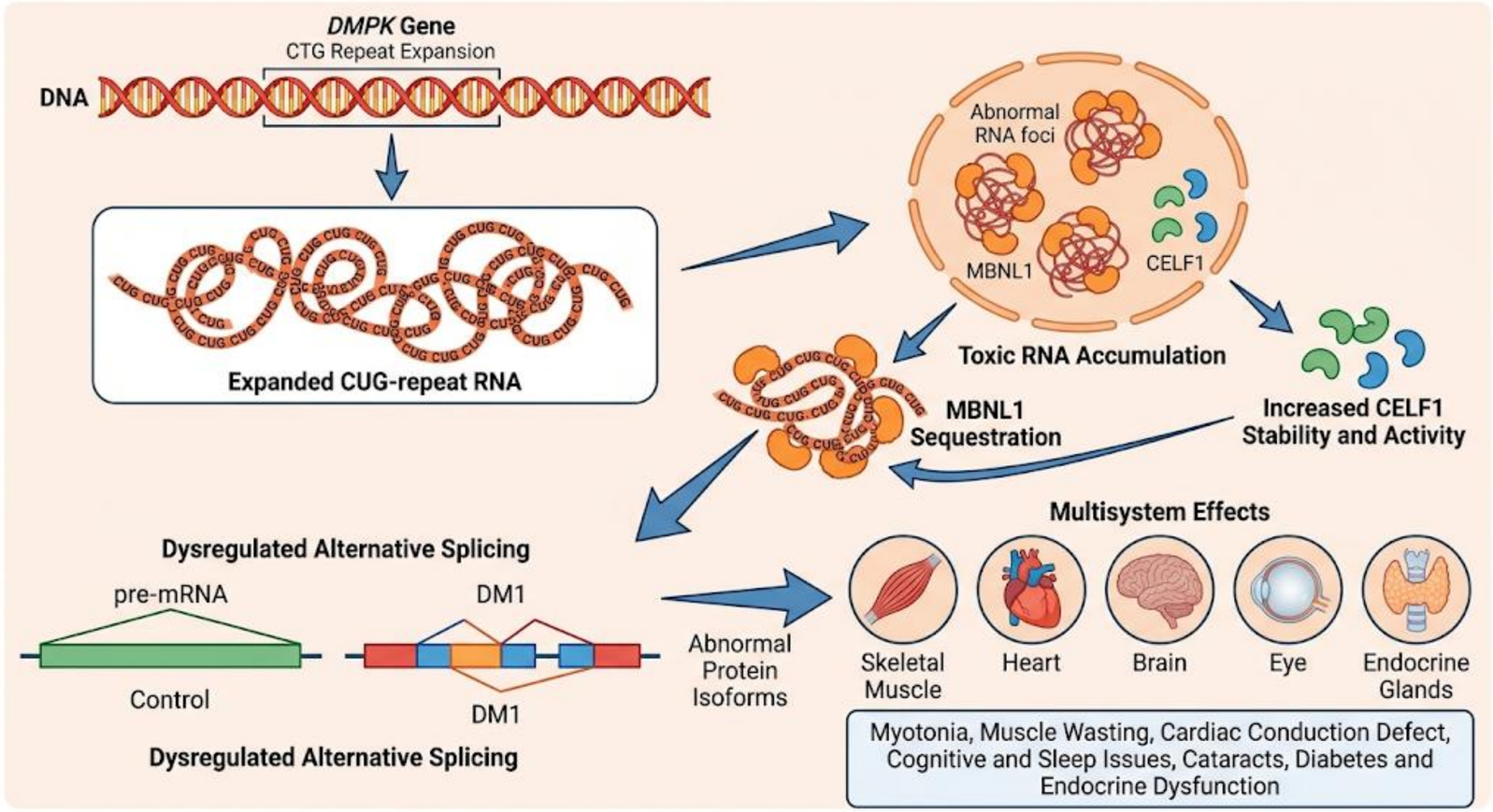


Image generated by Google Gemini

CTG repeat expansion in the DMPK gene

↓

Expanded CUG-repeat RNA
Abnormal RNA accumulates in the cell nucleus

↓

Sequestration of MBNL proteins and altered CELF1 activity
Disrupted regulation of RNA splicing and other RNA-processing pathways

Spliceopathy
MBNL-dependent RNAs from various genes are misspliced

Chloride channel RNA	→ myotonia
Insulin receptor RNA	→ diabetes
Calcium channel RNA	→ myopathy
Sodium channel RNA	→ arrhythmia

→ Multisystem disease

MBNL: Muscleblind-like proteins, a family of RNA-binding proteins involved in regulating RNA splicing.
CELF1: CUGBP Elav-like Family Member 1, an RNA-binding protein that regulates RNA processing.

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DM1 Affects the Whole Body

Cognitive impairment
Trouble concentrating, daytime sleepiness

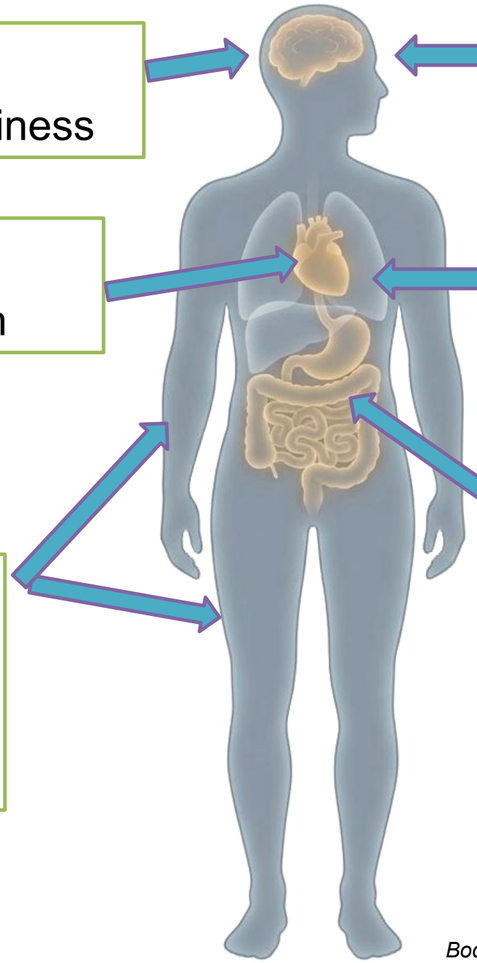
Cataracts
Early clouding of the eye lens

Arrhythmia
Irregular heart rhythm

Respiratory difficulties
Weak breathing muscles, disrupted sleep

Muscle weakness, atrophy, myotonia
Grip trouble, foot drop, gradual muscle loss

Gastrointestinal problems
Bloating, constipation, swallowing trouble



Body image generated by Google Gemini.

1.Myotonic Dystrophy Foundation. (2025). *Understanding DM1: From genetics to care* [Video]. YouTube. https://youtu.be/H2H2tr7jfag
2.The France Foundation. (2025). *Myotonic dystrophy type 1: Mechanism of disease* [Video]. YouTube. https://youtu.be/x_1dZlBXl88

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DM1 is an autosomal dominant genetic disorder

- **One copy is enough**

A person can develop DM1 if they inherit a disease-associated expanded CTG repeat in the DMPK gene from either parent.

- **Each pregnancy is an independent event**

If one parent has DM1, each child has a 50% chance of inheriting the expanded allele.

- **The expansion can change between generations**

The CTG repeat may become longer when passed from parent to child, which can lead to earlier onset or more severe disease.

This is called **anticipation** and is particularly associated with transmission from the mother in congenital DM1.

- **Repeat-length instability, variable clinical expression**

The number of CTG repeats can change when the expanded allele is transmitted. Siblings who both inherit the expanded allele may therefore have different repeat lengths.

Symptoms and severity can differ even among relatives with DM1. Repeat length is an important factor, but it does not perfectly predict an individual's symptoms or age of onset.

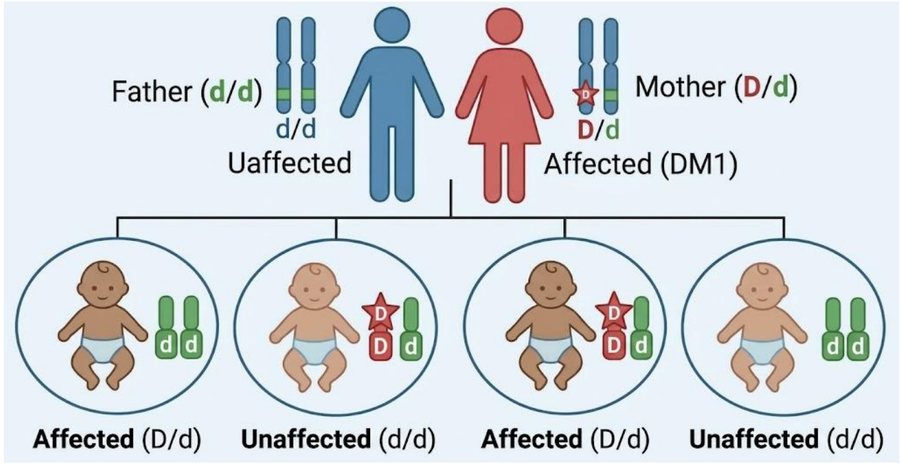
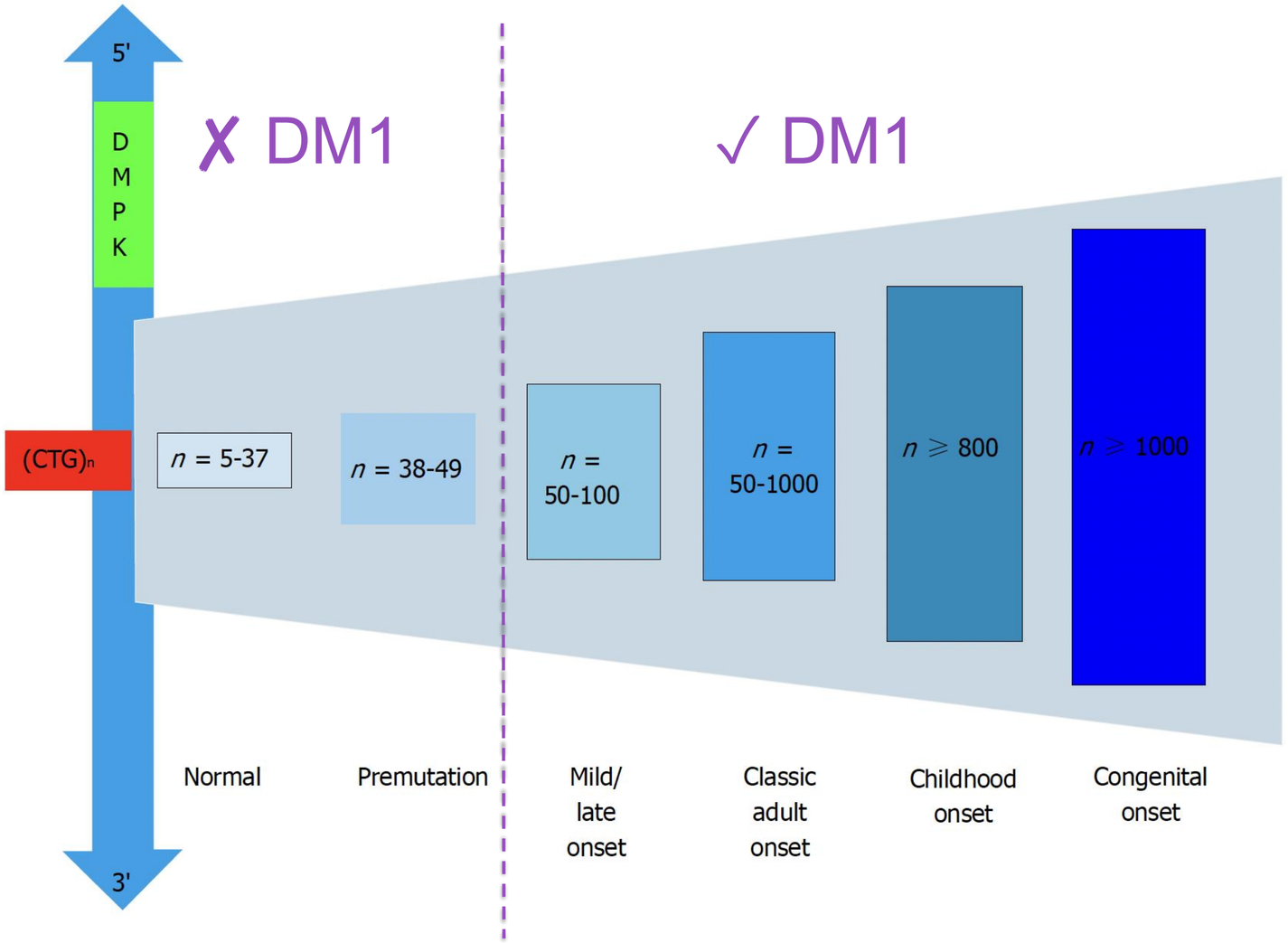


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1.Myotonic Dystrophy Foundation. (2025). *Understanding DM1: From genetics to care* [Video]. YouTube. https://youtu.be/H2H2tr7jfag
2.The France Foundation. (2025). *Myotonic dystrophy type 1: Mechanism of disease* [Video]. YouTube. https://youtu.be/x_1dZlBXl88

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While larger CTG expansions broadly correlate with earlier, more severe disease, individual progression remains unpredictable due to tissue-specific mosaicism



1.Ho G, Cardamone M, Farrar M. Congenital and childhood myotonic dystrophy: Current aspects of disease and future directions. *World J Clin Pediatr*. 2015;4(4):66-80. Published 2015 Nov 8. doi:10.5409/wjcp.v4.i4.66
2.Myotonic Dystrophy Foundation. (2025). *Understanding DM1: From genetics to care* [Video]. YouTube. https://youtu.be/H2H2tr7jfag

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Why one treatment may not benefit every patient equally?

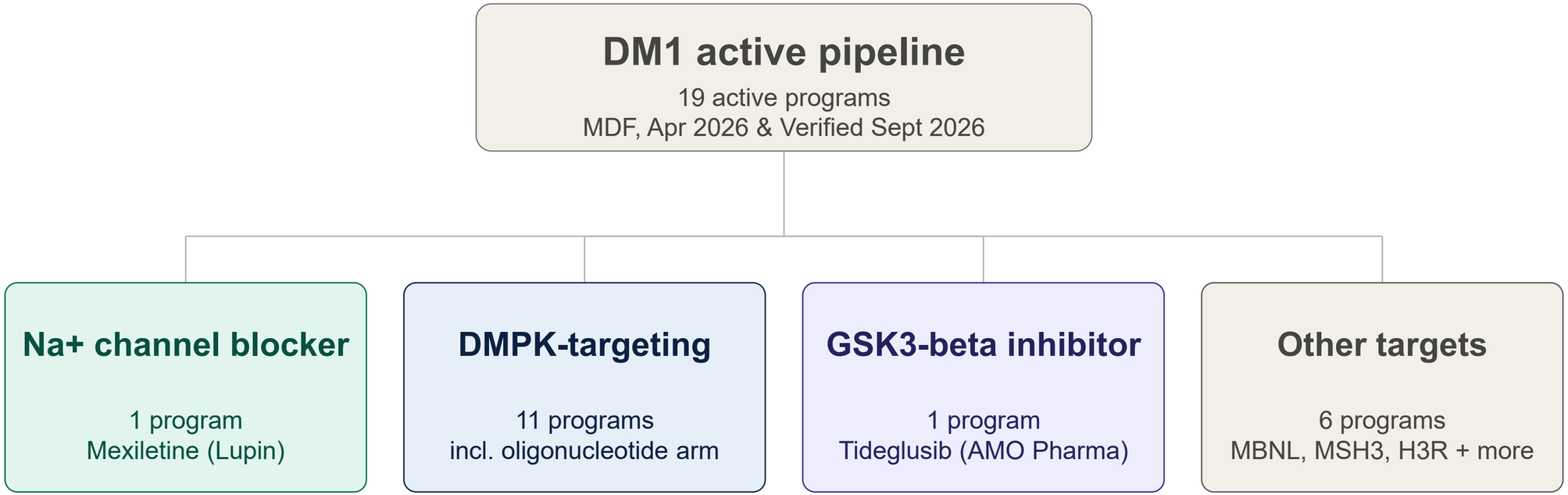
Multisystem nature is important for patient stratification and treatment-response prediction.

- Different patients with DM1 may have different dominant manifestations, even with the same underlying genetic disease.
- A treatment that improves muscle function may not improve cardiac or cognitive outcomes to the same extent.
- Treatment response should be assessed using endpoints relevant to the therapeutic target, such as muscle strength, myotonia, cardiac conduction, or respiratory function.
- Disease severity, CTG repeat instability, age, and tissue-specific RNA-processing effects may contribute to variability in treatment response.

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DM1 Therapeutics

DM1 active pipeline landscape



Source: MDF Drug Development Pipeline, April 2026 (myotonic.org). Status cross-checked against ClinicalTrials.gov and company disclosures, September 2026.

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Three distinct modalities target DM1 at different physiological levels

Symptomatic

Therapy:
Mexiletine

Focus:
Immediate relief of muscle stiffness

Disease-Modifying

Therapy:
Oligonucleotide

Focus:
RNA-level structural correction

Dual-pathway

Therapy:
Tideglusib

Focus:
Dual-pathway repair

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Treatment effects are hard to measure and compare

Comparing treatments is hard

- Trials report improvement on endpoints such as video hand opening time (vHOT).
- Each trial uses its own endpoint, time point, and population.
- Each drug is compared with its own placebo, not with other drugs.

vHOT varies widely

n = 591

Patients with baseline vHOT

5.9 ± 7.0 s

Baseline mean ± SD, median 3.9 s

0.4 ± 4.7 s

Mean change over 1 year (n = 270)

Same average shift, wide spread between patients

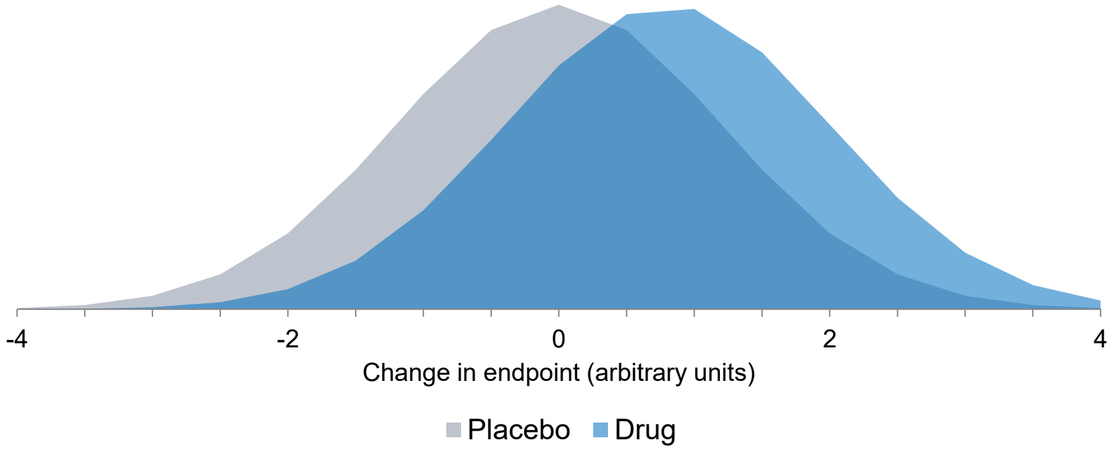


Illustration only. Not trial data.

Sources: Sansone et al., WMS 2025 (Dyne ACHIEVE poster); MDF, The Pathway to an Approvable Therapeutic; Leeuwenberg et al., Neurology 2026

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Your task: find what drives response

Identify

For each treatment arm, find the baseline features that drive response.

Beware

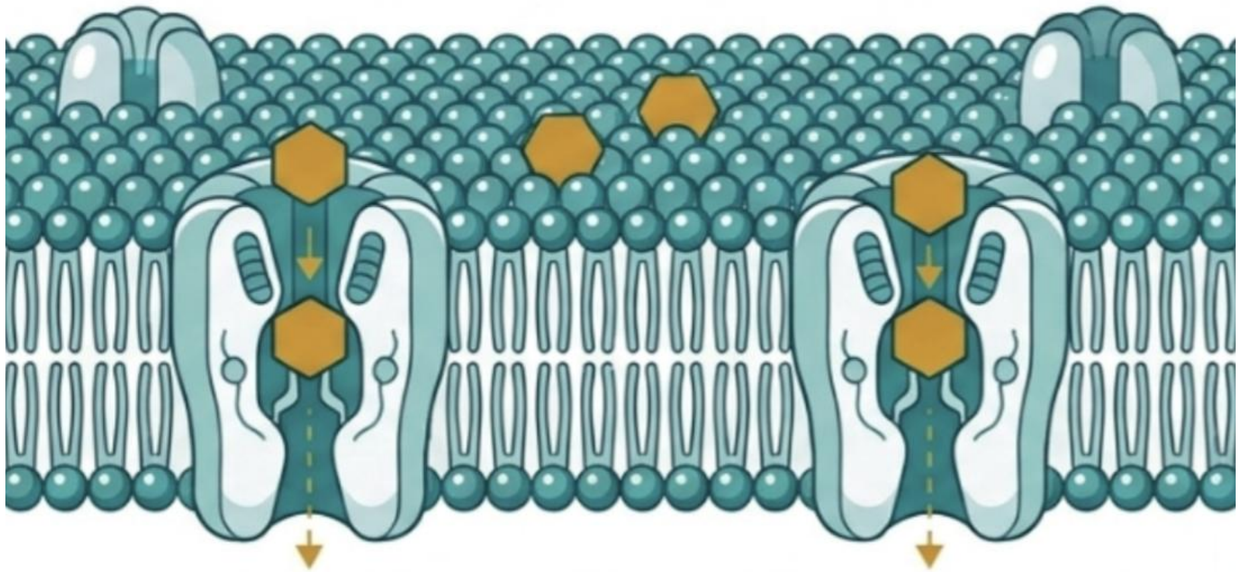
Not every covariate carries signal. Some look plausible but are noise.

Justify

Arms may differ. Explain why your features make **clinical** and statistical sense.

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Mexiletine relieves muscle stiffness via sodium channel blockade



Functions as a targeted sodium channel blocker.

- Directly **reduces myotonic discharges** in skeletal muscle, calming the hyperexcitability that causes prolonged muscle contractions.

Sources: De Bellis et al., Int J Mol Sci 2023.

How mexiletine response is assessed at 12 weeks



Outcome Endpoint

Metric: Change in myotonia severity at 12 weeks. Negative values indicate improvement.

In the clinic: Trials in myotonic disorders measure hand grip relaxation time and patient-reported stiffness.

Beyond myotonia: A 6-month placebo-controlled DM1 trial tested 6-minute walk distance and found no significant change.

Takeaway: Relief of myotonia does not always translate into functional gain.



Efficacy Considerations

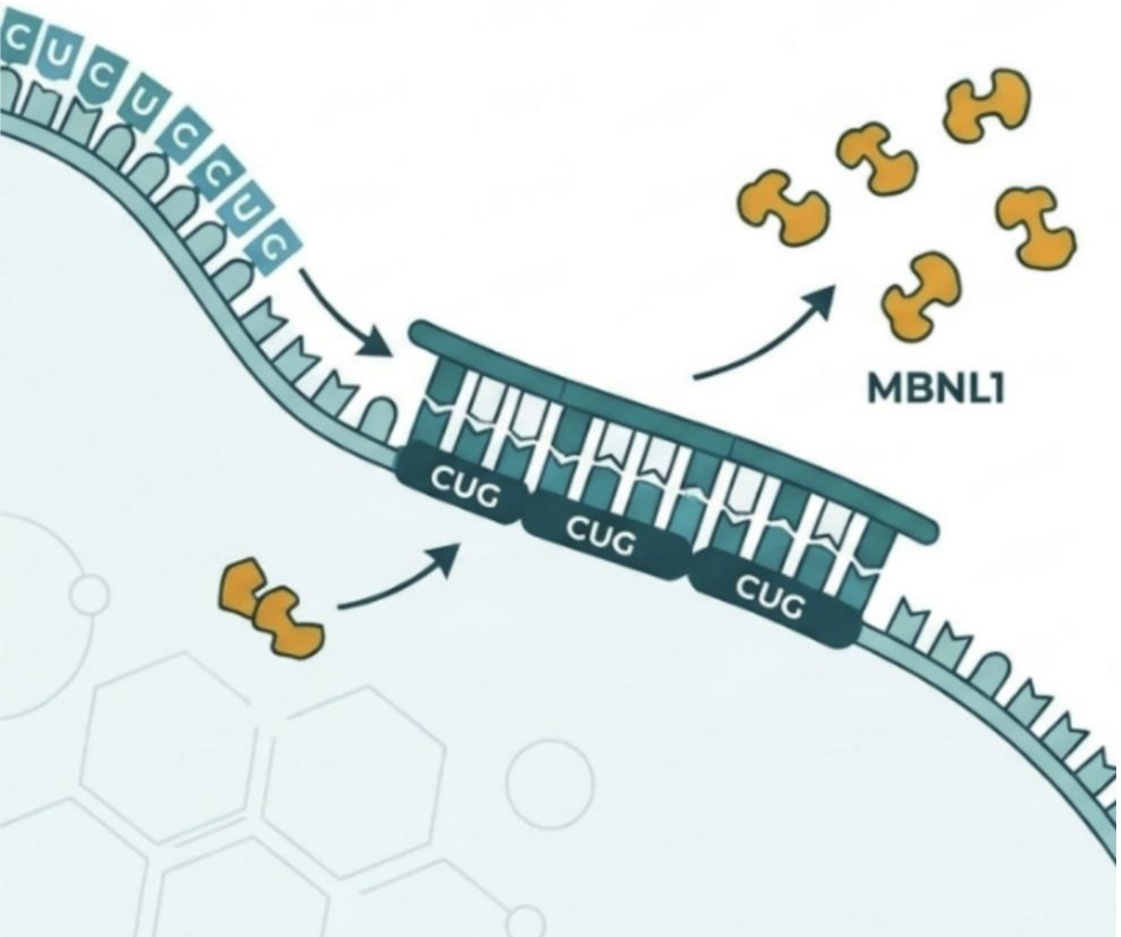
Efficacy: The drug acts on muscle excitability, so clinical presentation may matter.

Baseline: Some clinical features may be informative, while others may not.

Sources: Mexiletine in DM1 randomized trial, Neurology 2021; MYOMEX trial in non-dystrophic myotonia

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Oligonucleotides intercept the genetic messenger



An antisense oligonucleotide designed to target mutated DMPK mRNA.

- Reduces **toxic mutant RNA** and frees sequestered MBNL proteins, restoring normal cellular splicing and function.

Sources: Dyne ACHIEVE abstract, AAN 2025; Dyne corporate presentation

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How oligonucleotide response is assessed at 26 weeks



Outcome Endpoint

Metric: Change in Quantitative Muscle Testing (QMT) composite at 26 weeks. Positive values indicate improvement.

In the clinic: QMT is a quantitative muscle measure. Trials pair it with video hand opening time (vHOT) for myotonia.

Supporting measures: Timed walk tests, patient-reported health indices, and splicing biomarkers.

Note: vHOT is a common primary endpoint. This dataset uses QMT instead.



Efficacy Considerations

Efficacy: The drug acts on RNA, so molecular and clinical burden may both matter.

Baseline: Not every disease marker is equally informative

Sources: Dyne ACHIEVE data release, May 2024; Dyne corporate presentation (vHOT as primary endpoint)

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Tideglusib inhibits GSK3-beta

GSK3-beta inhibition



Modulation of Wnt/beta-catenin signaling and other kinase-regulated pathways

Regulation of protein synthesis and RNA-binding protein activity

Functions as an irreversible GSK3-beta inhibitor.

- GSK3-beta is a kinase overactive in DM1
- Simultaneously targets and corrects dysregulation in different pathways, uniquely addressing the multi-systemic nature of the disease.

Sources: REACH-CDM protocol (NCT03692312); GSK3-beta and CUGBP1 signaling in DM1, Int J Mol Sci 2020

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How tideglusib response is assessed at 26 weeks



Outcome Endpoint

Metric: Combined cognitive and motor improvement score (0 to 20) at 26 weeks.

In the clinic: The congenital DM1 trial (REACH-CDM) used the clinician-rated CDM1-RS as its primary endpoint.

Key secondary: Clinical Global Impression of Improvement (CGI-I), plus walk/run tests and caregiver-reported measures.

Result: REACH-CDM missed its primary endpoint but showed benefit on several functional assessments.



Efficacy Considerations

Efficacy: The target pathway is active in several tissues, so more than one domain may matter.

Baseline: Features from different domains may each contribute.

Sources: REACH-CDM statistical analysis plan (NCT03692312); NeurologyLive, September 2023

10-minute Break

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Overall Workflow I: Data, Arms & Split

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Overview of the dataset

The dataset is simulated data rather than real patient data.
Full details are available in “DM1_Data_Dictionary_Participant.xlsx”.

500
Simulated patients

18
Baseline predictor covariates

3
Treatment arms, one per patient

26
Total columns in the file

18 predictor covariates, by type

13
Numeric / continuous

ctg_repeats, somatic_mosaicism_index, age, age_at_onset, disease_duration, myotonia_severity, muscle_strength, pr_interval, cognitive_function, ck_level, insulin_resistance, albumin, bmi

5
Categorical

phenotype (3 levels), sex (2), family_history (3), cataracts (2), smoking_status (3)

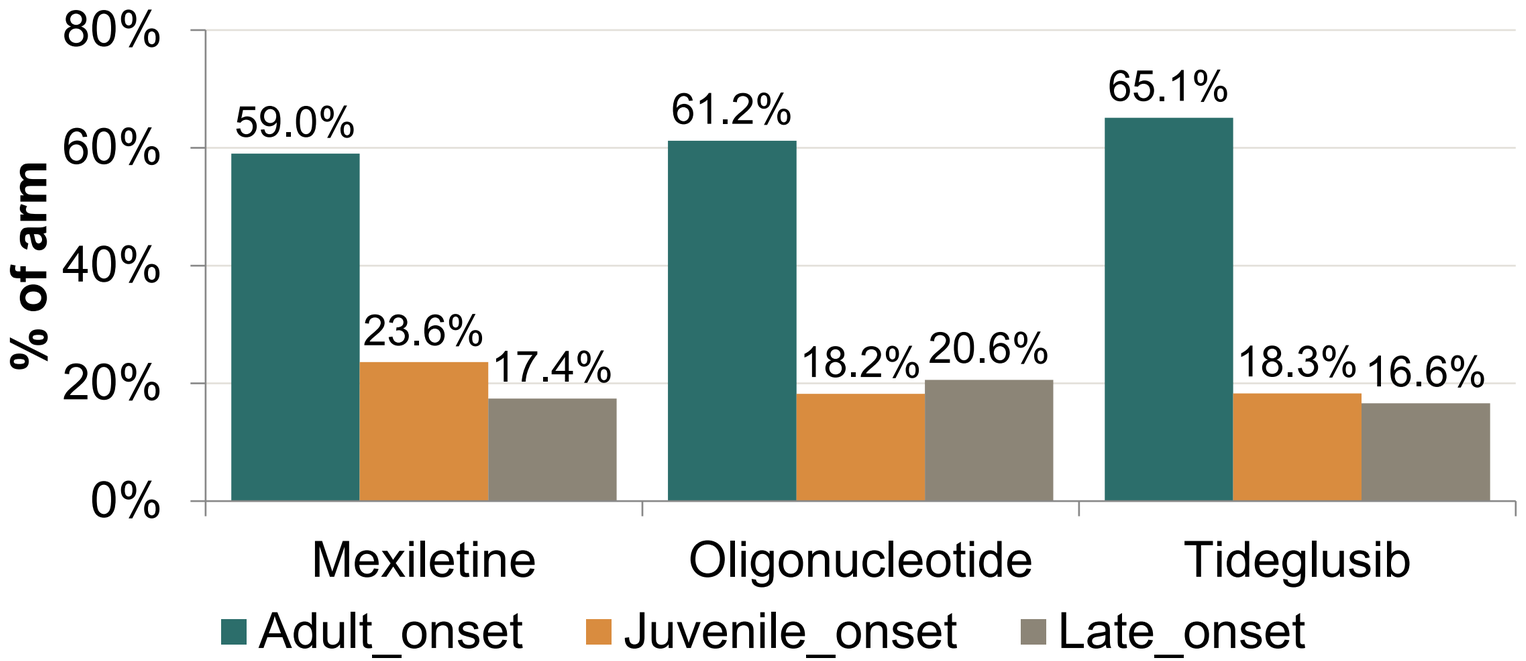
Plus, per patient

- 1 continuous outcome
- 1 binary responder label (responder/non-responder)

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Phenotypes are at similar proportions across arms

Share of each arm's patients by phenotype (n = 500)



Adult onset (n = 309, 61.8%)
Juvenile onset (n = 100, 20.0%)
Late onset (n = 91, 18.2%)

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Choose your outcome: continuous or binary, both are valid

Each arm has a continuous outcome and a binary responder label derived from it

Arm	Continuous outcome	Binary label
Mexiletine	**myotonia_change_12wk** Change in myotonia severity (delayed muscle relaxation) at Week 12. Negative = improvement.	**responder_phenotype_mex** Responder / Non-responder
Oligonucleotide	**qmt_change_26wk** QMT change at Week 26. Positive = improvement.	**responder_phenotype_oligo** Responder / Non-responder
Tideglusib	**tideglusib_composite** Combined symptoms' improvement score (0 to 20) at Week 26.	**responder_phenotype_tidge** Responder / Non-responder

Regression on the continuous outcome

Suggested metrics: RMSE, MAE, R^2

Classification on the binary label

Suggested metrics: ROC AUC, recall, precision

Either choice is valid. State which outcome you model and why.

QMT, Quantitative Muscle Testing.

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Overall Workflow II: Pipeline & Deliverables

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The pipeline runs from understanding to fair evaluation

Each step protects the next one from bias.

1

Understand

Explore structure, distributions, and relationships.

2

Split/Cross Validation

Hold out a test set before any modeling.

3

Preprocess

Encode, scale, and clean only as the method requires. Fit on train only.

4

Select

Match the method to the size and shape of the data.

5

Train

Fit and tune using the training set only.

6

Evaluate

Score once on the held-out test set.

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Step 1: Explore the data before choosing any model

Understand what is in each arm before deciding what to model. Once you have split, explore the training data only.

Structure

Each patient belongs to one arm. The outcome exists only for that arm.

Distributions

Check ranges, skew, and outliers for the 13 numeric covariates.

Covariate variation

Summarize every covariate within each arm. Note how much each one varies.

Relationships

Look for correlated covariates and for class balance in the responder label.

Step 2: Split once, per arm, before any modeling

Your team decides how to split. These four rules keep it unbiased.

Training set

The model learns here, like practice problems.

Test set

Final check, like an exam.

1. **Split each arm separately**
   Outcomes are arm-specific.

2. **Stratify by outcome***
   Keep the outcome distribution similar in training and test.

3. **Fix the random seed**
   The split stays reproducible.

4. **Lock the test set**
   Use it only for the final evaluation.

*Use the responder label directly. For a continuous outcome, group it into bins first.

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Step 3: Preprocess only what the chosen method needs

Prepare the data only as your method requires. One rule protects your results.

Avoid leakage: train and test stay separate

1. **Split first**
   Hold out the test set before any preprocessing.
2. **Fit on train only**
   Learn any preprocessing from the training data.
3. **Apply to test**
   Reuse the same fitted steps. Never refit on test data.

Preprocessing means any cleaning or transformation your method requires.

Step 4: Select the method that fits the data

Let the data characteristics drive the choice.

Data trait	Ask	Implication
Sample size	How many patients per arm?	Simpler or penalized models suit small samples
Covariates	How many, and of what types?	Plan encoding and variable selection
Relationships	Correlations or interactions?	Choose methods that can handle them
Outcome type	Continuous or binary?	Match metrics to the outcome

Covariate modeling

Machine learning

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You are free to choose your platform, language, and AI tools

The workflow matters more than the software.

Platform and language

- Use any platform or programming language you prefer.
- R, Python, and other tools are all welcome.
- Follow the same six steps regardless of tool.

AI use clarification

- LLM chatbots and AI agents are welcome.
- You must understand every step of your final analysis.
- Report which AI tools you used and for which purpose.
- **You remain accountable for all results.**

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Before you submit: check every item.

Each item supports a scoring criterion. **Upload to your team SharePoint folder by the end of Tue Oct 13.**

Slides tell one story: workflow, key covariates, interpretation, conclusion.	Communication, Clinical
Code files are complete and reproduce your results.	Technical
Split, noise handling, and metrics are explained and justified.	Technical
Key covariates are separated from noise.	Clinical
AI tools used are reported, with the purpose of each.	All criteria

Presentation: 5 min plus 3-min Q&A, Wed Oct 14, 9:00 AM.
At least one member of your team needs to help present and no limit on how many people will participate in the presentation.

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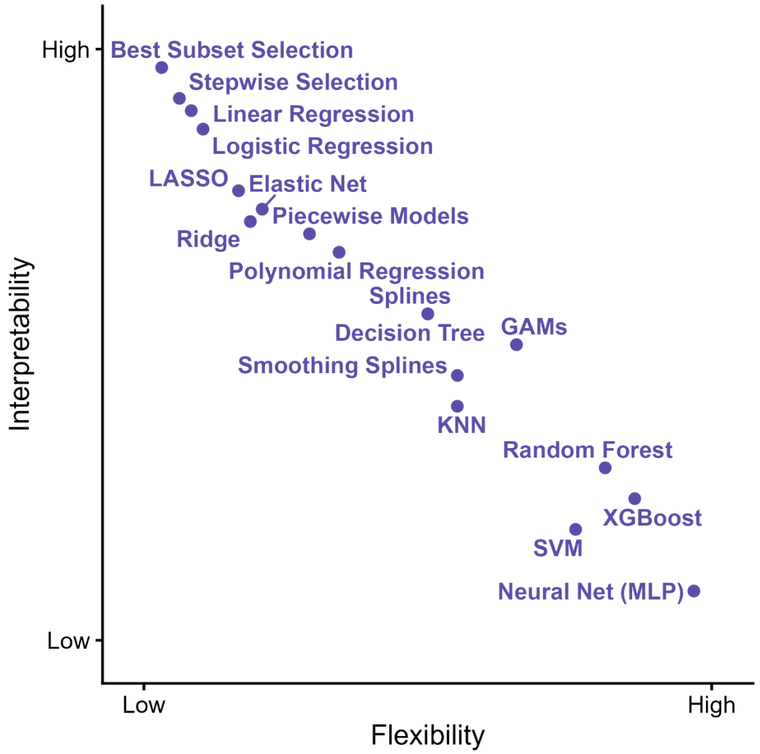
Covariate Model Methodologies

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Covariate Search vs. Machine Learning: Where's the Line?

Both use the same data but ask different questions

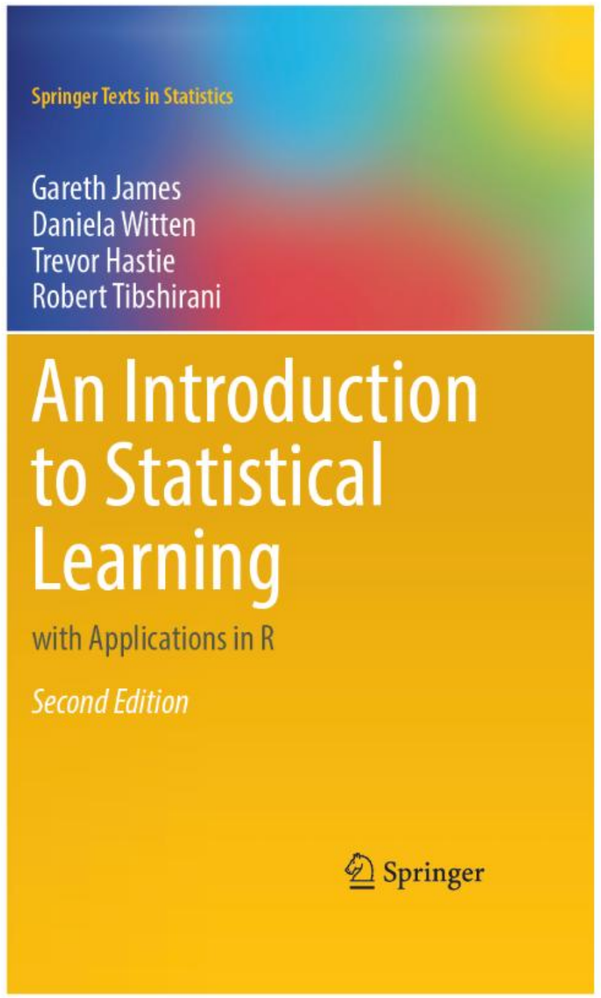
	Covariate Search	Machine Learning
Core question	Which variables matter and how?	Can we predict the outcome accurately?
Priority	Interpretability & inference	Predictive performance
Output	Coefficients, selected variables	Predictions, feature importance
Risk	Oversimplification	Black box

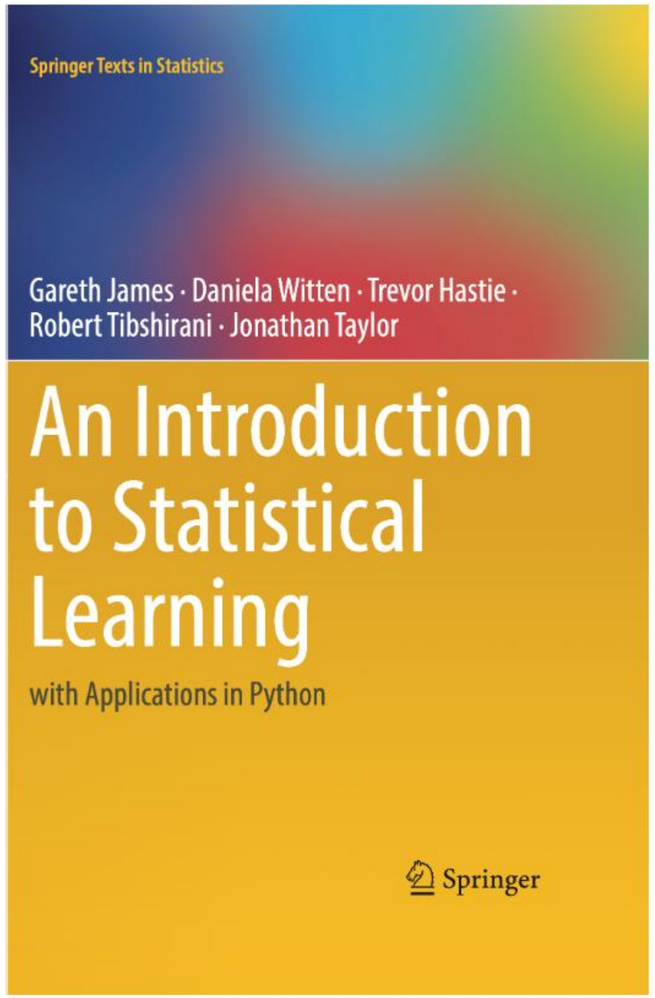


💡 *Many methods live in both worlds: LASSO selects covariates AND predicts. GAMs are interpretable AND flexible. The line is blurry by design*

Start here: ISLR textbook (free)

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Springer Texts in Statistics
Gareth James
Daniela Witten
Trevor Hastie
Robert Tibshirani
An Introduction to Statistical Learning
with Applications in R
Second Edition
Springer


Springer Texts in Statistics
Gareth James · Daniela Witten · Trevor Hastie · Robert Tibshirani · Jonathan Taylor
An Introduction to Statistical Learning
with Applications in Python
Springer

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Linear Regression: the starting point

Fits a straight line between each covariate and the outcome; the slope is the effect size

Analogy

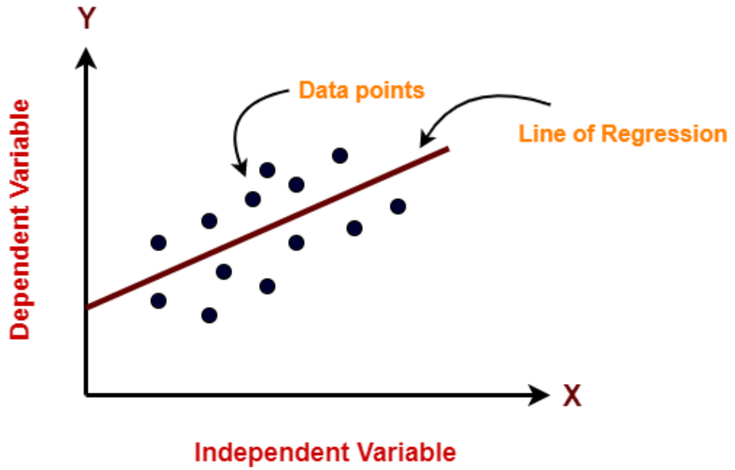
A ruler laid through a cloud of points. The angle of the ruler tells you how much the outcome changes per unit of the covariate.

For our dataset

A natural first step for the continuous efficacy outcome. Coefficients directly tell you which patient characteristics are associated with treatment response magnitude.

Watch out

Assumes a linear relationship and normally distributed residuals. Sensitive to outliers and won't capture non-linear covariate effects.




```
# Fit linear regression

model <- lm(efficacy_outcome ~ age + sex +
baseline_score + ..., data = dm1_data)
summary(model)  # coefficients, p-values, R²
```

Start here: CRAN lm() documentation

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Stepwise Selection: letting the data choose

Automatically adds or removes covariates one at a time based on a model fit criterion (AIC/BIC)

Analogy

Packing a suitcase by trying items one at a time. Forward adds things that help, backward removes things that don't, and both does a mix of both

For our dataset

Useful for narrowing down number of covariates to a manageable set for each treatment arm without manually testing every combination.

Watch out

Results can be unstable. A slightly different dataset can yield a very different selected set.

Don't treat the final model as "the truth."

```
library(MASS)
full_model <- lm(efficacy_outcome ~ ., data = dm1_data)
null_model <- lm(efficacy_outcome ~ 1, data = dm1_data)

# Forward
step_fwd <- stepAIC(null_model, scope = list(upper =
full_model), direction = "forward")
# Backward
step_bwd <- stepAIC(full_model, direction = "backward")
# Both
step_both <- stepAIC(full_model, direction = "both")
```

Start here: MASS package documentation

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Best Subset Selection: exhaustive search

Fits every possible combination of covariates and picks the best model at each size

Analogy

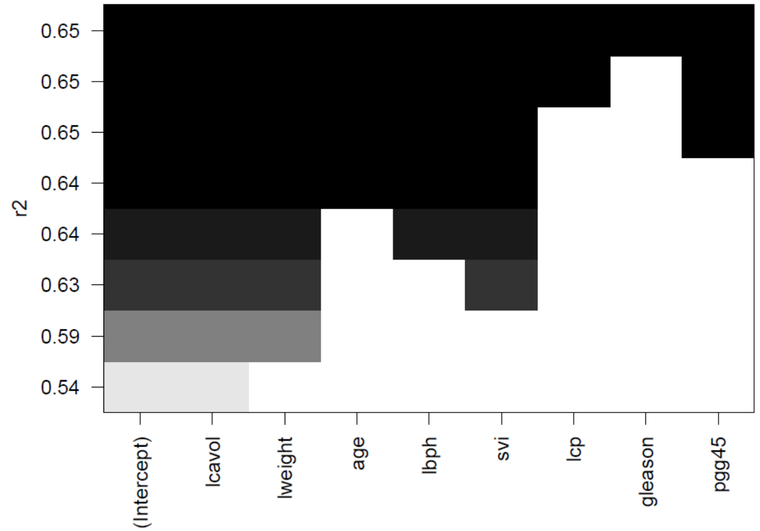
Instead of trying items one at a time, you try every possible combination of items for your suitcase and pick the best-fitting set.

For our dataset

With our dataset size, best subset is still computationally feasible and gives you the globally optimal set of covariates of any given size

Watch out

Becomes computationally expensive beyond ~20-25 covariates. Still subject to overfitting if you don't use a proper selection criterion (prefer BIC over R^2).




```
library(leaps)

subset_model <- regsubsets(efficacy_outcome ~ ., data =
dm1_data, nvmax = 10)
summary_sub <- summary(subset_model)
```

Start here: leaps package on CRAN

ACoP 2026

LASSO, Ridge & Elastic Net: penalized regression

Add a penalty to the regression to shrink coefficients — the shape of the penalty determines whether covariates are kept, shrunk, or zeroed out

Analogy

A budget constraint on your model. Each covariate costs something to include. The type of budget rule determines whether you cut some expenses completely or spread cuts evenly

For our dataset

All three work for both continuous efficacy outcome and binary responder classification. Ridge is valuable when many correlated covariates each contribute modestly — it stabilizes rather than arbitrarily drops them. LASSO gives a sparse solution. Elastic Net combines both.

Watch out

All three require cross-validation to choose λ. LASSO can arbitrarily drop one of two highly correlated covariates — clinically meaningful variables may disappear not because they're unimportant but because another variable is correlated with them.

Method	Penalty	Coefficient behavior
LASSO	$\lambda\sum\lvert\beta_j\rvert$	Some set exactly to zero (sparse solution)
Ridge	$\lambda\sum\beta_j^2$	Shrunk toward zero, never exactly zero
Elastic Net	$\lambda_1\sum\lvert\beta_j\rvert + \lambda_2\sum\beta_j^2$	Some set to zero, correlated ones shrunk together

```
library(glmnet)
# All three methods, same function only alpha changes

cv_ridge <- cv.glmnet(X, y, alpha = 0) # Ridge
cv_lasso <- cv.glmnet(X, y, alpha = 1) # LASSO
cv_enet  <- cv.glmnet(X, y, alpha = 0.5)# Elastic Net

# For binary outcome, just add family argument
cv_lasso_bin <- cv.glmnet(X, y_binary, alpha = 1,
family = "binomial")
```

Start here: glmnet vignette

Add more: Group LASSO, Adaptive LASSO, SCAD, MCP, …

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Polynomial Regression & Piecewise Models: bending the line

Extends linear regression by allowing curved or segmented relationships between a covariate and the outcome.

Analogy

Instead of one straight road, you allow the road to bend (polynomial) or change direction at specific points (piecewise).

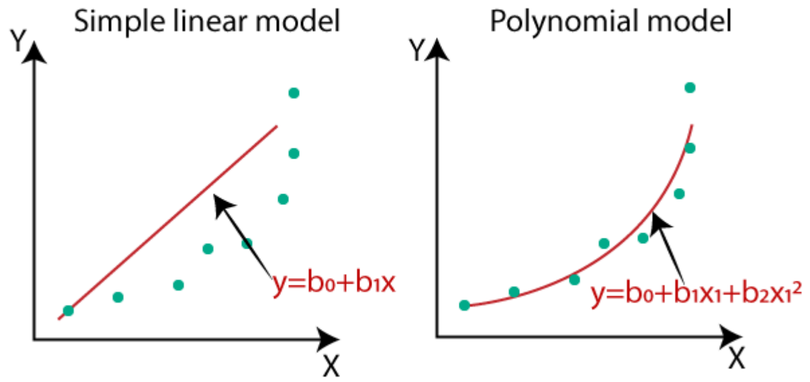
For our dataset

Patient characteristics such as disease duration, muscle strength scores, or biomarker levels may not relate linearly to treatment response. Polynomial terms and piecewise models allow us to capture these curved relationships without fully abandoning the interpretability of regression

Watch out

High-degree polynomials can oscillate wildly at the edges of your data (Runge's phenomenon). Piecewise models require careful choice of breakpoints (knots).

Start here: ISLR Chapter 7 — Moving Beyond Linearity




```
# Polynomial regression (degree 2)
model_poly <- lm(efficacy_outcome ~ poly(age, 2), data
= dm1_data)
# Piecewise / step function
dm1_data$age_cut <- cut(dm1_data$age, breaks = c(0, 30,
50, 70, 100))
model_piece <- lm(efficacy_outcome ~ age_cut, data =
dm1_data)
```

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Splines & Smoothing Splines: flexible curves with control

Fits smooth curves through the data by joining polynomial pieces at knots, with a smoothness penalty to avoid overfitting

Analogy

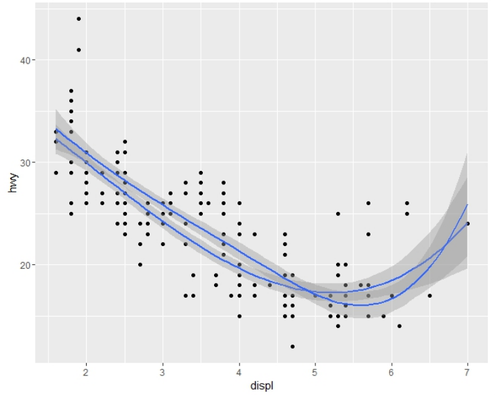
A flexible ruler (a physical spline) bent to pass smoothly through a set of points. Flexible enough to follow the data, stiff enough not to wiggle wildly.

For our dataset

Allows continuous covariates like age or baseline score to have smooth, non-linear effects on efficacy without manually specifying the shape.

Watch out

Choosing the number and placement of knots matters for regression splines. Smoothing splines handle this automatically via a smoothness penalty but are harder to interpret.




```
library(splines)
# Regression spline with 3 knots
model_spline <- lm(efficacy_outcome ~ bs(age, knots =
c(30, 50, 65)), data = dm1_data)
# Smoothing spline
smooth_fit <- smooth.spline(dm1_data$age,
dm1_data$efficacy_outcome, cv = TRUE)
plot(smooth_fit)
```

Start here: splines package documentation

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Generalized Additive Models (GAMs): smooth effects for every covariate

Extends linear regression by allowing each covariate to have its own smooth, non-linear effect — all added together.

Analogy

A linear model where instead of a straight ruler for each covariate, you use a flexible curve. Each covariate gets its own shape, and you add all the shapes together.

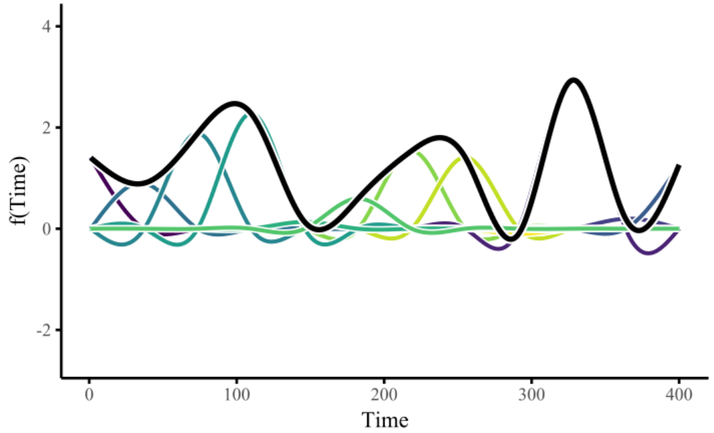
For our dataset

Handles both continuous and categorical covariates, works for both continuous efficacy outcomes and binary responder classification, and produces interpretable smooth effect plots per covariate.

Watch out

More complex to interpret than linear regression. Smoothness is controlled by the k parameter. Too small and you miss real patterns, too large and you overfit.

Start here: mgcv package & GAM tutorial by Noam Ross




```
library(mgcv)
# GAM for continuous outcome
gam_model <- gam(efficacy_outcome ~ s(age) +
s(baseline_score) + sex + treatment, data = dm1_data,
method = "REML")
summary(gam_model)
# For binary outcome (responder/non-responder)
gam_logit <- gam(responder ~ s(age) + s(baseline_score)
+ sex, data = dm1_data, family = binomial, method =
"REML")
```

2026

PCA: simplifying the covariate space

Combines correlated covariates into a smaller set of uncorrelated components that capture the most variance in the data.

Analogy

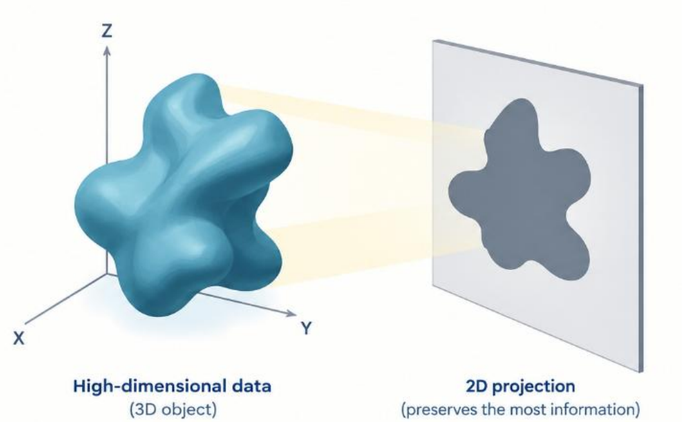
Instead of describing a person by height, arm length, and leg length separately (all correlated), you summarize them with one "body size" component.

For our dataset

Useful as a pre-processing step when patient characteristics are correlated. Can reveal whether patients cluster naturally before any outcome is considered, useful for understanding dataset structure.

Watch out

PCA components are linear combinations of original covariates. They can be hard to interpret clinically. It is unsupervised, so it doesn't know about the treatment outcome when it creates components.




```
# Run PCA on numeric covariates
pca_data <- dm1_data[, numeric_covariates]
pca_result <- prcomp(pca_data, scale. = TRUE)
# Variance explained
summary(pca_result)
# Biplot, see which covariates drive which components
biplot(pca_result, scale = 0)
```

Start here: StatQuest: Principal Component Analysis (PCA), Step-by-Step

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Where to learn more

One start-here page per topic. The full list is in the resource spreadsheet.

An Introduction to Statistical Learning
ISLR textbook (free)

Linear Regression in R
CRAN lm() documentation

Stepwise Selection in R
MASS package documentation

Best Subset Selection in R
leaps package on CRAN

LASSO, Ridge & Elastic Net
glmnet vignette

Polynomial Regression & Piecewise Models
ISLR Chapter 7 — Moving Beyond Linearity

Splines & Smoothing Splines
splines package documentation

Generalized Additive Models (GAMs)
mgcv package & GAM tutorial by Noam Ross

Principal Component Analysis (PCA)
StatQuest: Principal Component Analysis (PCA), Step-by-Step

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15-minute Break

Coffee will be available!

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Machine Learning Model Methodologies



Logistic Regression: a weighted scorecard

It gives each covariate a weight, adds them up, and converts the total to a probability.

Analogy

A clinic risk score. Each factor adds or removes points, and the total sets the risk.

For our dataset

A transparent baseline. Each weight shows how a covariate pushes the odds of response up or down.

Watch out

It assumes each covariate acts in a straight line, so it can miss complex interactions.



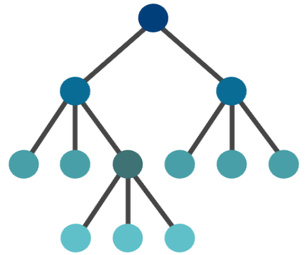

```
from sklearn.linear_model import LogisticRegression

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
print(model.coef_)  # one weight per covariate
```

Start here: scikit-learn.org/stable/modules/linear_model.html#logistic-regression

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Decision Tree: a clinic flowchart



It asks a chain of yes/no questions about covariates. Each path ends in a prediction.

Analogy

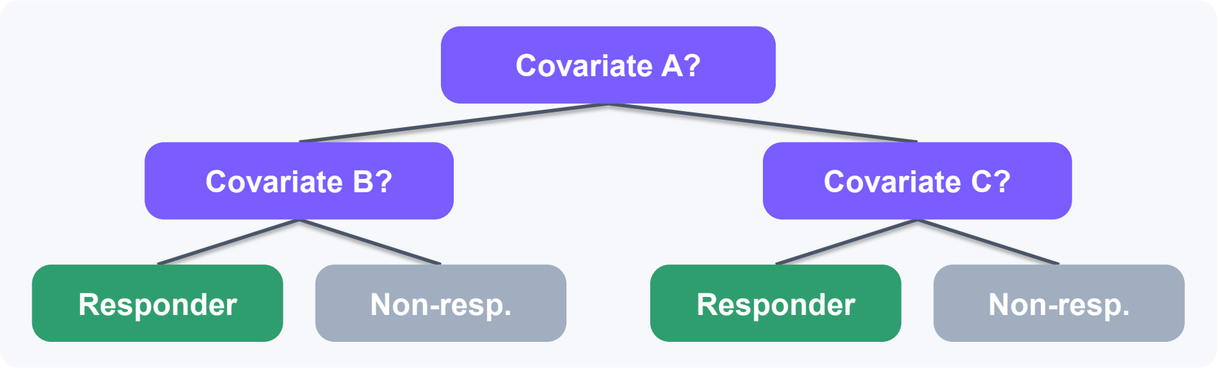
A triage flowchart. Each question splits patients into smaller, more similar groups.

For our dataset

Shows cutoffs on covariates that separate responders from non-responders. Easy to read aloud.

Watch out

One deep tree memorizes noise. Limiting the depth keeps it general.




```
from sklearn.tree import DecisionTreeClassifier

model = DecisionTreeClassifier(max_depth=3)
model.fit(X_train, y_train)
```

Start here: scikit-learn.org/stable/modules/tree.html

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Random Forest: a panel of trees

It grows many different trees on random slices of the data. The trees then vote.

Analogy

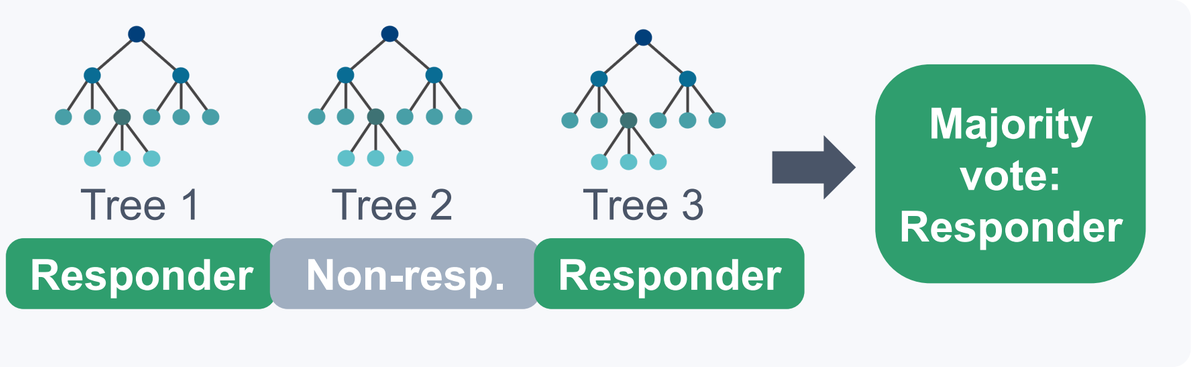
Asking a panel of clinicians instead of one. Individual errors tend to cancel out.

For our dataset

Captures interactions between covariates without us specifying them. It also ranks covariates.

Watch out

Harder to read than one tree. The vote hides the individual rules.




```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier(n_estimators=300)
model.fit(X_train, y_train)
```

Start here: scikit-learn.org/stable/modules/ensemble.html

ACoP 2026

Support Vector Machine: the widest margin

It finds the boundary that separates two groups with the widest possible gap.

Analogy

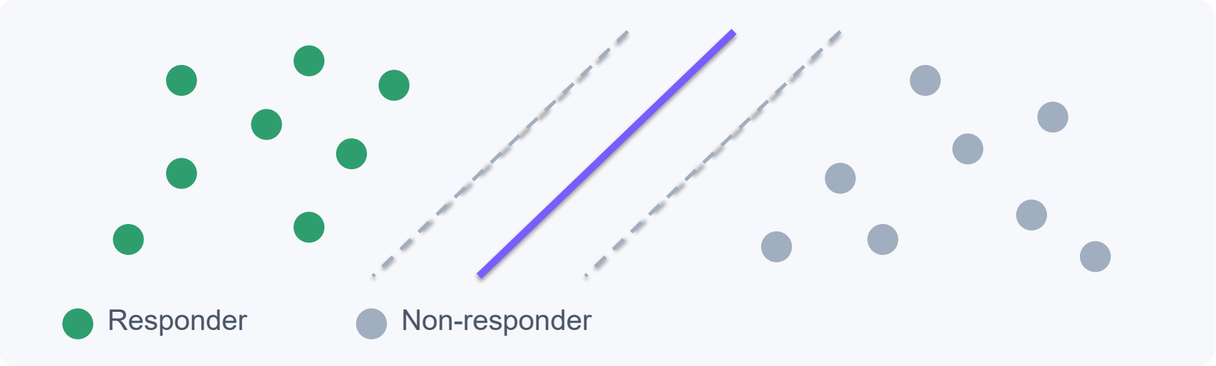
Drawing the fairest line between two crowds, as far as possible from both.

For our dataset

Can draw curved boundaries when responders and non-responders overlap in complex ways.

Watch out

Hard to explain to clinicians. It gives a boundary, not simple rules.




```
from sklearn.svm import SVC

model = SVC(kernel="rbf", probability=True)
model.fit(X_train, y_train)
```

Start here: scikit-learn.org/stable/modules/svm.html

ACoP 2026

K-Nearest Neighbors: patients like you

It finds the K most similar past patients and copies their majority outcome.

Analogy

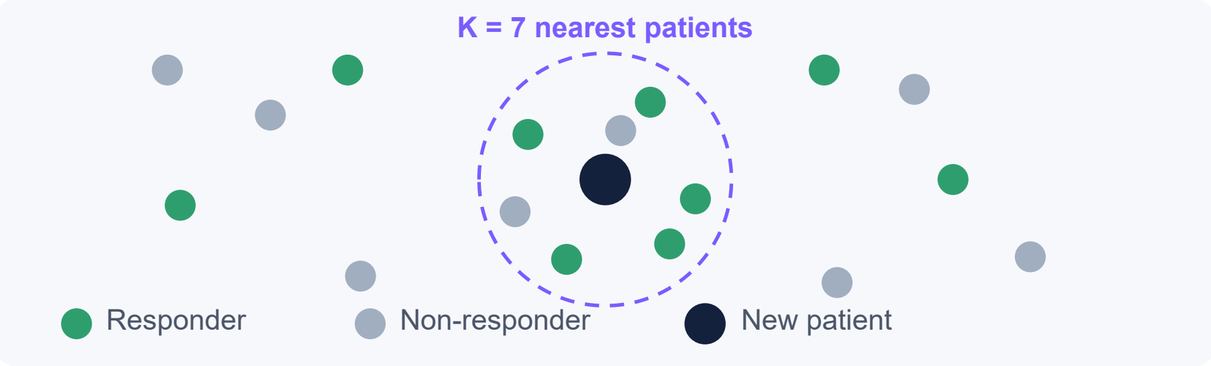
"Patients like you did well on this drug." The closest matches decide the verdict.

For our dataset

Very intuitive for clinicians. It relies on how similar two patient profiles are.

Watch out

Noise covariates dilute similarity and can mislead the neighbors.




```
from sklearn.neighbors import KNeighborsClassifier

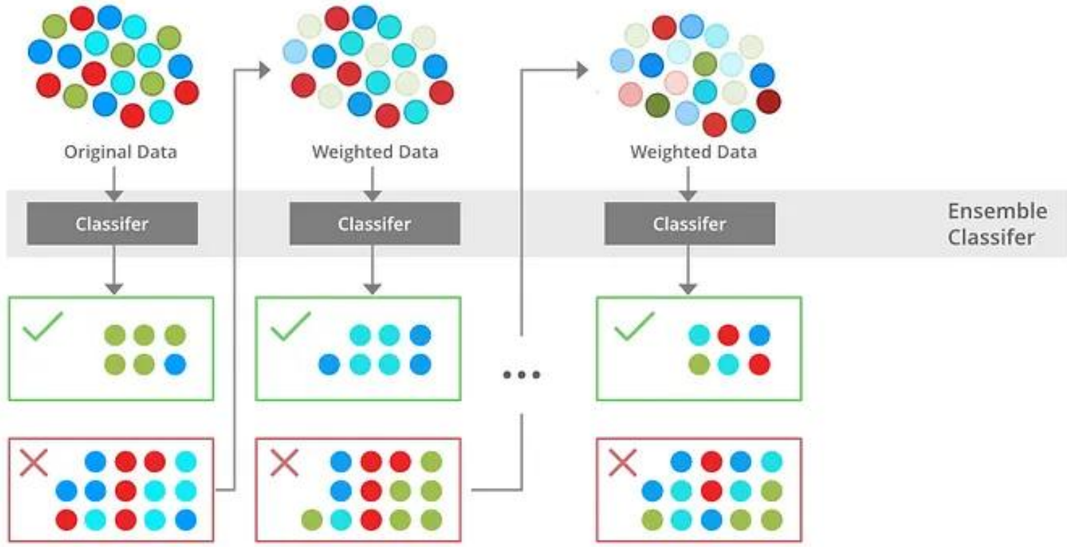
model = KNeighborsClassifier(n_neighbors=7)
model.fit(X_train, y_train)
```

Start here: scikit-learn.org/stable/modules/neighbors.html

ACoP 2026

XGBoost: learning from mistakes

It builds small trees one after another.
Each new tree fixes the errors of the last.



Source: https://medium.com/@techynilesh/xgboost-algorithm-explained-in-less-than-5-minutes-b561dcc1ccee

Each tree is small and weak. Together they are strong.

Analogy

A student who reviews only the questions they got wrong, round after round.

For our dataset

Often strong on tabular clinical data. It handles mixed covariate types and interactions.

Watch out

Many settings to tune. With few patients, it can overfit quickly.

```
from xgboost import XGBClassifier

model = XGBClassifier(n_estimators=200, max_depth=3)
model.fit(X_train, y_train)
```

Start here: xgboost.readthedocs.io/en/stable/tutorials/model.html

ACoP 2026

Neural Network (MLP): learned combinations

A hidden layer combines covariates into new features. The output layer makes the prediction.

Analogy

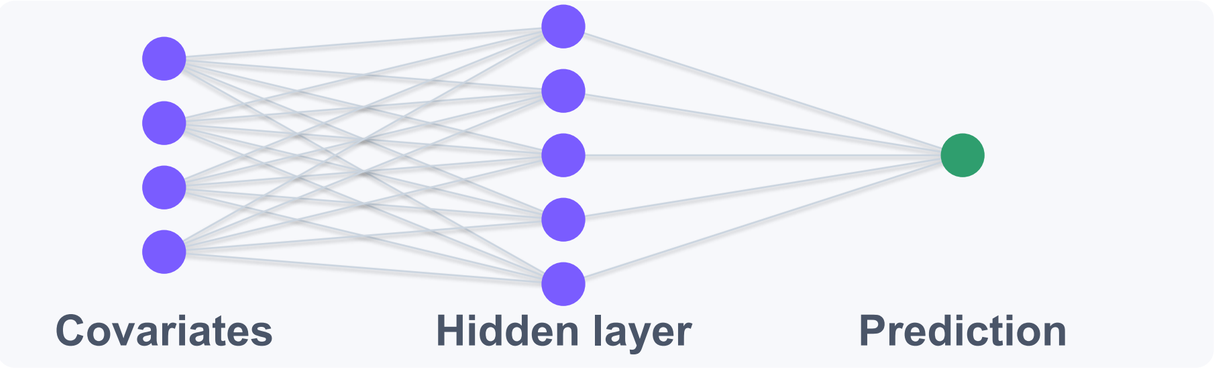
A team where each member summarizes the inputs and passes the summary forward.

For our dataset

Use a deliberately tiny network. It could test whether deep learning adds value on small data.

Watch out

The least transparent model. Small arms limit what it can learn.




```
from sklearn.neural_network import MLPClassifier

model = MLPClassifier(hidden_layer_sizes=(16,))
model.fit(X_train, y_train)
```

Start here: scikit-learn.org/stable/modules/neural_networks_supervised.html

ACoP 2026

Hyperparameter tuning: finding the best settings

Every model has settings, such as tree depth. Grid search tries combinations and keeps the best.

Analogy

Adjusting oven temperature and time until the bake works.

For our dataset

Gives every model a fair chance. Grids stay small because each arm has few patients.

Watch out

A large grid on a small arm mostly finds noise. Keep grids coarse.




```
from sklearn.model_selection import GridSearchCV
depth = [2, 3, 5, None]
leaf = [1, 5, 10]
grid = {"max_depth": depth, "min_samples_leaf": leaf}
search = GridSearchCV(model, grid, scoring="roc_auc")
search.fit(X_train, y_train)
print(search.best_params_)
```

Start here: scikit-learn.org/stable/modules/grid_search.html

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Permutation importance: which covariates did the model use?

Shuffle one covariate. If predictions get worse, the model relied on it.

Analogy

Removing one ingredient from a recipe and tasting the difference.

For our dataset

Ranks covariates for each arm. It helps separate likely drivers from noise.

Watch out

Correlated covariates can share credit, which hides each one's true role.




```
from sklearn.inspection import permutation_importance

result = permutation_importance(
    model, X_test, y_test, n_repeats=20)
print(result.importances_mean)
```

Start here: scikit-learn.org/stable/modules/permutation_importance.html

ACoP 2026

SHAP: explaining each prediction

It splits one patient's prediction into a contribution from every covariate.

Analogy

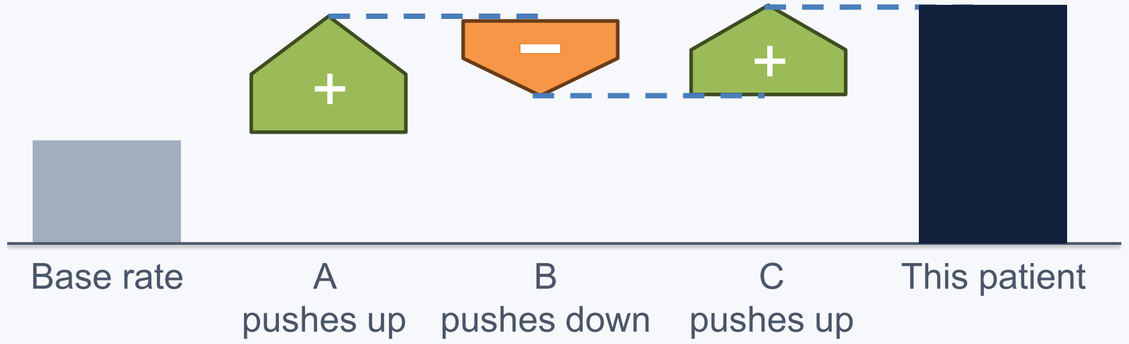
An itemized receipt. Each covariate adds to or subtracts from the final total.

For our dataset

Shows which covariates matter and in which direction, for each arm's model.

Watch out

It explains the model, not biology. Importance does not prove causation.




```
import shap

explainer = shap.TreeExplainer(model)
shap_values = explainer.shap_values(X_test)
shap.summary_plot(shap_values, X_test)
```

Start here: shap.readthedocs.io

ACoP 2026

Where to learn more

One start-here page per topic. The full list is in the resource spreadsheet.

Choosing a model
scikit-learn.org/stable/machine_learning_map.html

Logistic regression
scikit-learn.org/stable/modules/linear_model.html

Trees and forests
scikit-learn.org/stable/modules/tree.html

SVM and KNN
scikit-learn.org/stable/modules/svm.html

XGBoost
xgboost.readthedocs.io/en/stable/tutorials/model.html

Neural network
scikit-learn.org/stable/modules/neural_networks_supervised.html

Tuning
scikit-learn.org/stable/modules/grid_search.html

Importance and SHAP
shap.readthedocs.io

Interpretation book (free)
christophm.github.io/interpretable-ml-book

Code-first blog tutorials
machinelearningmastery.com

ACoP 2026

Assessment & Evaluation Metrics



Continuous outcomes: how far off are the predictions?



Orange lines are prediction errors. Each metric summarizes them differently.

RMSE

Square root of the mean squared error. Penalizes large prediction errors more heavily. Lower is better.

MAE

Mean absolute error, average magnitude of prediction errors, in the outcome's original units. Lower is better.

R^2

coefficient of determination; measures how much variation the model explains relative to predicting the test-set mean.

1 - Perfect prediction,

0 - No improvement in squared error over predicting the mean outcome of the evaluation set.

<0 - Worse squared-error performance than that mean-based reference

- RMSE and MAE are in the units of the outcome, so they cannot be compared across arms.
- Test-set R^2 can be negative when a model performs worse than the baseline of predicting the mean of the observed test-set outcomes for every observation.
- If an outcome was scaled, convert predictions back to the original units before computing RMSE and MAE. Remember that the direction of improvement differs by arm, so report it alongside the error.

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Classification starts with four outcomes

Each test patient lands in one cell. Every classification metric is built from these counts.

	Predicted responder	Predicted non-responder
Actual responder	**3** TP: responder found	**1** FN: responder missed
Actual non-responder	**2** FP: false alarm	**14** TN: non-responder found

Illustrative example: 20 test patients, 4 true responders. Not a result. Responder is the positive class.

Recall `TP / (TP + FN)`

Of the true responders, how many did we find?
Here 3 of 4 is 75%.

Precision `TP / (TP + FP)`

Of the predicted responders, how many were right?
Here 3 of 5 is 60%.

Accuracy `(TP + TN) / all`

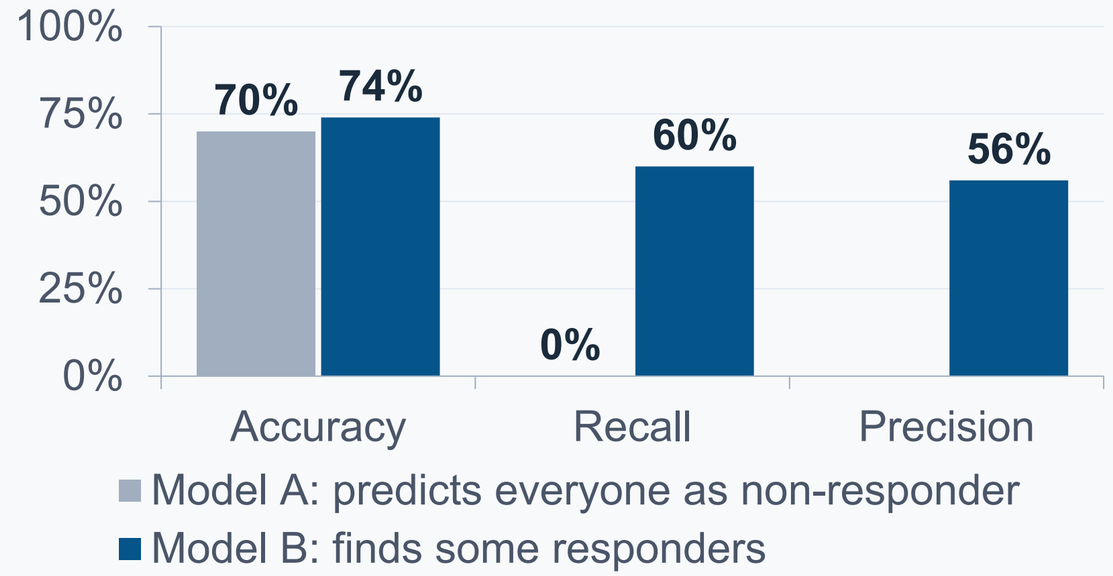
Share of all patients classified correctly.
Here 17 of 20 is 85%.

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Why accuracy alone can mislead

When non-responders are the majority, a model that finds no responders can still look accurate.

Two hypothetical models on the same 100 test patients



Illustrative example: 30 responders and 70 non-responders. Model A has no precision because it never predicts a responder. Not a result.

Report recall and precision

Show both for the responder class, not only overall accuracy.

Check the class mix

Count responders and non-responders in your test set before you read accuracy.

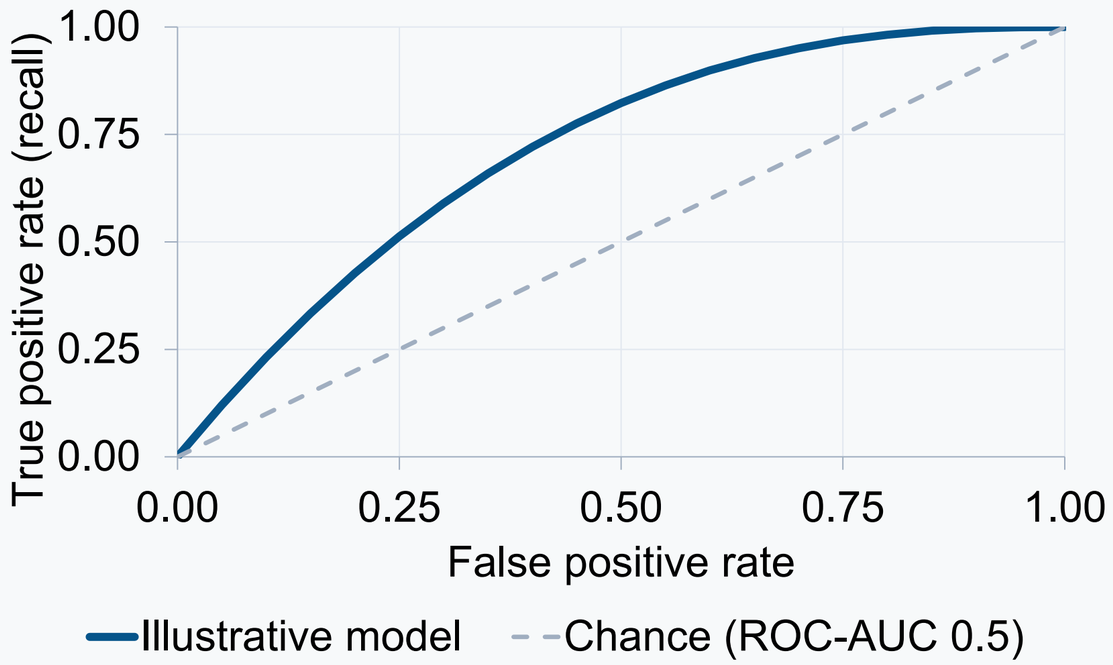
Lean on ROC-AUC

It ignores the class mix. A model with no skill scores 0.5.

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ROC-AUC: how well does the model rank patients?

It asks whether true responders score higher than non-responders, across every cutoff.



Reading it

0.5 means random guessing and 1.0 means perfect ranking. A higher curve is better.

Cutoffs trade off

A lower cutoff finds more responders but adds false alarms. ROC-AUC needs no single cutoff.

Not the PK AUC

ROC-AUC scores classification. It has nothing to do with drug exposure.

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Build one scoreboard per arm

Example layout. Put covariate and ML models in one table, and fill it from the same test patients.

Method	Type	ROC-AUC	Recall	Precision	Accuracy
Stepwise selection	Covariate	…	…	…	…
LASSO / Elastic Net	Covariate	…	…	…	…
Random Forest	ML	…	…	…	…
XGBoost, untuned	ML	…	…	…	…
XGBoost, tuned	ML	…	…	…	…

Scoreboard rules

Keep untuned and tuned rows side by side.

Add a bootstrap interval if time allows.

Do not over-read small gaps. With only a few dozen test patients per arm, one patient moves the scores noticeably.

- Repeat the table for each arm.
- For a continuous outcome, swap the four metric columns for RMSE, MAE and R^2.

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Compare the covariates, not only the scores

Two methods can score alike and point to different covariates. Report what each one selected.

Covariate models

Report the covariates each method keeps. Stepwise lists selected terms. LASSO and Elastic Net list non-zero coefficients.

ML models

Report an importance ranking. Permutation importance and SHAP show which covariates the model used.



Illustration only: generic labels, not results.

Compare

Which covariates appear across methods? Where methods disagree, explain why using the data and clinical plausibility.

Why it matters

Agreement across methods builds confidence. A covariate only one method picks needs a second look.

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Before you present: evaluation checklist

Each item supports the Technical score. Judges look for metrics that fit the outcome.

Do

1. Match the metrics to the outcome type.
2. Test every method on the same patients.
3. Report results per arm.
4. Pair accuracy with recall and precision.
5. State your cutoff for predicted labels.

Watch out

Constant predictions

A penalty that drops every covariate gives ROC-AUC of 0.5.

Small test sets

Scores are noisy, so avoid ranking methods on tiny gaps.

Tuning

It does not always help on small data. Show untuned and tuned results.

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Where Teams Might Show Genuine Innovation

Standard Technical Execution

Generally applies the provided methods correctly and validates results properly.

- Correct use of covariate or ML methods
- Proper train / test separation
- Reasonable interpretation

Genuine Innovation

Typically goes further than technically correct execution.

- Connects findings to plausible clinical or biological mechanisms
- Proposes thoughtful patient stratification or trial-design implications
- Communicates through clear, decision-relevant visuals rather than black-box output

Thank you!

Q&A

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Participant Team Formation (subject to change)

Team 1	Team 2	Team 3	Team 4	Team 5
Euibeom Shin	Adhithi Lakshmikanthan	Yaping Liu	Segun Oke	Behnoosh Bonakdar Yazdi
Luong Vuong	Philip Delff	Isabel Figueroa	Minseok Kim	Kapil Gadkar
Vijay Ivaturi	Sanjay Kharche	Mark Peterson	Narayan Cheruvu	Seth Berry
Ming Gao	Parsshava Mehta	Fatma Kir	Izumi Hamada	Frano Mihaljevic
Wenzhe Ma	Olayide Oladoyinbo	Ron Keizer	Yongjun Hu	Ya-Feng Wen
Bei Zhao	Shayne Watson	Stephanie Kollmann	Sripriya Moorthy	

Team 6	Team 7	Team 8	Team 9	Team 10
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Mina Park	Leyanis Rodriguez Vera	Mark Sale	Phil Bowsher	Tim Liang
Csaba Katai	Pradeep Natarajan	Kun Yang	Wangui Mbuguiro	Gang Sun
Lan Gao	Priyata Kalra	Pratik Bhagunde	Inez Lam	Xiaoping Zhang
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